Lesson 25 Counting Functions

1. Let $A = \{1,2,3\}$ and $B = \{1,2,3,4,5\}$.
   a) How many relations are there from A to B?
   b) How many relations are there from B to A?
   c) How many functions are there from A to B?
   d) How many functions are there from B to A?
   e) How many one-to-one functions are there from A to B?
   f) How many one-to-one functions are there from B to A?
   g) How many onto functions are there from A to B?
   h) How many onto functions are there from B to A?
   i) How many bijections are there from A to B?
   j) How many bijections are there from B to A?
2. Repeat Question 1, for relations and functions from $A = \{1, 2, 3, 4\}$ to itself.
3. Let $A = \{1, 2, \ldots, n\}$, $B = \{0, 1\}$, and $C = \{0, 1, 2\}$
   a) How many functions $f: A \to B$ are there are such that $f(a) = 0$ for exactly k elements a?
   b) How many functions $f: A \to C$ are there are such that $f(a) = 0$ for exactly k elements a?
4. Let A and B be finite sets with $|A| = |B|$. Prove that $f: A \to B$ is one-to-one, then f is onto.